

MM 263 (Real Analysis)

Practice Problem Set - 3

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1. If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous, then show that
 - (a) there exist $a, b \in [0, 1]$ such that $a - b = \frac{1}{2}$ and $f(a) - f(b) = \frac{1}{2}(f(1) - f(0))$.
 - (b) there exist $a, b \in [0, 1]$ such that $a - b = \frac{1}{3}$ and $f(a) - f(b) = \frac{1}{3}(f(1) - f(0))$.
2. Let $\alpha > 1$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $|f(x)| \leq |x|^\alpha$ for all $x \in \mathbb{R}$. Show that f is differentiable at 0.
3. Let $f(x) = x^2|x|$ for all $x \in \mathbb{R}$. Examine the existence of $f'(x)$, $f''(x)$ and $f'''(x)$, where $x \in \mathbb{R}$.
4. Give an example (with justification) of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is differentiable only at 2.
5. Let $f(x) = x^3 + x$ and $g(x) = x^3 - x$ for all $x \in \mathbb{R}$. If f^{-1} denotes the inverse function of f and if $(g \circ f^{-1})(x) = g(f^{-1}(x))$ for all $x \in \mathbb{R}$, then find $(g \circ f^{-1})'(2)$.
6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x) - f(y) \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Show that f is a constant function.
7. Find the number of (distinct) real roots of the following equations.
 - (a) $x^2 = \cos x$
 - (b) $e^{2x} + \cos x + x = 0$
 - (c) $x^{13} + 7x^3 - 5 = 0$
8. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable such that $f(0) = f(1) = 0$. Show that there exists $c \in (0, 1)$ such that $f'(c) = f(c)$.
9. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable such that $f(0) = 0$ and $f'(x) > f(x)$ for all $x \in \mathbb{R}$. Show that $f(x) > 0$ for all $x > 0$.
10. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function such that $f(x) \neq 0$ for all $x \in [a, b]$. Show that there exists $c \in (a, b)$ such that $\frac{f'(c)}{f(c)} = \frac{1}{a-c} + \frac{1}{b-c}$.
11. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable such that $f(0) = 0$ and $f(1) = 1$. Show that there exist $c_1, c_2 \in [0, 1]$ with $c_1 \neq c_2$ such that $f'(c_1) + f'(c_2) = 2$.
12. $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$ and $f(1) = 1$. Show that there exist $a, b \in (0, 1)$ with $a \neq b$ such that $\frac{1}{f'(a)} + \frac{1}{f'(b)} = 2$.

13. Let $f : [a, b] \rightarrow \mathbb{R}$ be twice differentiable and let $f(a) = f(b) = 0$ and $f(c) > 0$, where $c \in (a, b)$. Show that there exists $\xi \in (a, b)$ such that $f''(\xi) < 0$.
14. Using Taylor's theorem, show that
- $|\sqrt{1+x} - (1 + \frac{x}{2} - \frac{x^2}{8})| \leq \frac{1}{2}|x|^3$ for all $x \in (-\frac{1}{2}, \frac{1}{2})$.
 - $1 - \frac{x^2}{2!} + \frac{x^4}{4!} > \cos x > 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$ for all $x \in (0, \pi)$.
 - $x - \frac{x^3}{3!} < \sin x < x - \frac{x^3}{3!} + \frac{x^5}{5!}$ for all $x \in (0, \pi)$.
15. Evaluate the following limits.
- $\lim_{x \rightarrow 0} (\frac{1}{\sin x} - \frac{1}{x})$
 - $\lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}}}{x}$
 - $\lim_{x \rightarrow \infty} x(\log(1 + \frac{x}{2}) - \log \frac{x}{2})$
 - $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}-e}}{x}$
 - $\lim_{x \rightarrow \infty} \frac{2x + \sin 2x + 1}{(2x + \sin 2x)(\sin x + 3)^2}$
16. (a) Find $f : [0, 1] \rightarrow \mathbb{R}$ such that $|f|$ is integrable but f is not integrable.
 (b) Find $f : [0, 1] \rightarrow \mathbb{R}$ such that f^2 is integrable but f is not integrable.
17. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. If there is a partition P of $[a, b]$ such that $L(f, P) = U(f, P)$, then show that f is a constant function.
18. Let $f(x) = \begin{cases} x & \text{if } x \in [0, 1] \cap \mathbb{Q}, \\ 0 & \text{if } x \in [0, 1] \cap (\mathbb{R} \setminus \mathbb{Q}). \end{cases}$
 Examine whether f is Riemann integrable on $[0, 1]$. Also, find $\int_0^1 f$, if it exists (in $\mathbb{R}$).
19. Show that (a) $\frac{\pi^2}{9} \leq \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{x}{\sin x} dx \leq \frac{2\pi^2}{9}$. (b) $\frac{\sqrt{3}}{8} \leq \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx \leq \frac{\sqrt{2}}{6}$.
20. Let $f : [0, 1] \rightarrow \mathbb{R}$ and $f(x) \geq 0$ for all $x \in [0, 1]$.
- If f is integrable, show that $\int_0^1 f(x) dx \geq 0$.
 - If f continuous and $\int_0^1 f(x) dx = 0$. Show that $f(x) = 0$ for all $x \in [0, 1]$.
 - Give an example of an integrable function f on $[0, 1]$ such that $f(x) \geq 0$ for all $x \in [0, 1]$ and $\int_0^1 f(x) dx = 0$ but $f(x_0) \neq 0$ for some $x_0 \in [0, 1]$.
21. If $f : [0, 1] \rightarrow \mathbb{R}$ is Riemann integrable, then find $\lim_{n \rightarrow \infty} \int_0^1 x^n f(x) dx$.

22. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then show that there exists $c \in [a, b]$ such that $\int_a^b f(x) dx = (b - a)f(c)$.
(This result is called the mean value theorem of Riemann integrals.)
23. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous and let $g(x) \geq 0$ for all $x \in [a, b]$. Show that there exists $c \in [a, b]$ such that $\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx$.
(This result is called the generalized mean value theorem of Riemann integrals.)
24. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let $g(x) = \int_0^x (x - t)f(t) dt$ for all $x \in \mathbb{R}$. Show that $g''(x) = f(x)$ for all $x \in \mathbb{R}$.
25. Let $f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \\ 0 & \text{if } 1 < x \leq 2, \end{cases}$ and let $F(x) = \int_0^x f(t) dt$ for all $x \in [0, 2]$.
Is $F : [0, 2] \rightarrow \mathbb{R}$ differentiable? Justify.